1st REVIEW

Distributed Video Anomaly Detection in Large-Scale Surveillance Systems Using HDFS and Cluster-Visualized Federated CNN–LSTM Models

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KONERU LAKSHMAIAH EDUCATION FOUNDATION

GROUP NUMBER:1

Name of Candidate	Role
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ABSTRACT

Air travel involves several disconnected processes, including check-in, boarding-pass access, flight tracking, baggage monitoring, airport navigation, and assistance requests, often causing delays, confusion, and unnecessary dependence on airport counters. The **Airline & Airport Mobile App for Passenger Self-Service** proposes a unified mobile platform that brings these essential services together in one seamless application. Passengers can complete digital check-in, select seats, access QR-based boarding passes, track flight and baggage status, receive real-time gate and boarding notifications, navigate airport facilities, and request services such as special assistance. The system combines a user-friendly mobile interface with a secure backend and centralized database to provide personalized and efficient passenger services. By connecting airline and airport services within a single platform, the proposed solution aims to reduce airport congestion, minimize waiting time, improve information accessibility, and enhance the overall passenger experience. The application ultimately provides a smarter **"booking-to-boarding" digital journey**, demonstrating how self-service technology can make modern air travel more convenient, connected, and efficient.

INTRDOUCTION

- Air travel involves multiple processes such as **check-in, boarding, baggage handling, flight updates, and airport navigation**.
- Passengers often depend on **counters and information screens**, leading to delays and inconvenience.
- Our project provides a **single mobile platform** for essential airline and airport services.
- Passengers can **check in, access digital boarding passes, track flights and baggage, navigate the airport, and receive real-time alerts**.
- The goal is to create a **faster, smarter, and more convenient passenger journey** while reducing dependency on manual services.

EXISTING SYSTEM

- Passengers currently use **separate airline apps, airport websites, kiosks, and service counters** for different services.
- **Check-in, boarding passes, baggage tracking, and flight information** are often handled through different systems.
- Airport navigation usually depends on **signboards, information screens, or staff assistance**.
- Flight and gate changes may not always be communicated in a **single, centralized platform**.
- Passengers may experience **long queues, information gaps, delays, and difficulty accessing services** during peak hours.
- There is limited integration between **airline services and airport facilities** in one unified passenger application.
- These limitations create the need for a **single, intelligent self-service platform** for a smoother travel experience.

PROBLEM

- Air passengers often depend on **multiple platforms, airport counters, kiosks, and information screens** to access essential services.
- The lack of a **unified self-service platform** makes processes such as check-in, baggage tracking, flight updates, and airport navigation inconvenient.
- **Long queues and manual assistance** can increase waiting time, especially during peak travel periods.
- Passengers may miss important **gate changes, boarding updates, or baggage information** due to fragmented communication.
- Navigating unfamiliar airports can also be difficult without **real-time location and facility guidance**.
- Therefore, there is a need for a **single, user-friendly mobile application** that integrates airline and airport services to provide passengers with a faster, smarter, and seamless travel experience.

SOLUTION(Contribution)

- Develop a **unified mobile application** that brings airline and airport services into one platform.
- Enable passengers to complete **digital check-in, seat selection, and boarding-pass generation** without visiting a counter.
- Provide **real-time flight, gate, and boarding notifications** to keep passengers updated.
- Implement **baggage tracking** with a clear status timeline from check-in to collection.
- Provide an **interactive airport map** for finding gates, security, lounges, restaurants, restrooms, and other facilities.
- Allow passengers to request **special assistance and airport services** directly through the app.
- Use a **personalized dashboard** to display upcoming flights, boarding countdowns, alerts, and travel information.
- Integrate a secure **backend and database** to manage passenger, flight, booking, and baggage information.
- The solution aims to **reduce queues and waiting time, improve information accessibility, and provide a seamless booking-to-boarding experience**.

Objectives

- **Develop a unified mobile platform** integrating essential airline and airport passenger services.
- Enable **fast and convenient self-check-in** with digital seat selection and boarding-pass generation.
- Provide **real-time flight, gate, boarding, and baggage updates**.
- Implement **smart airport navigation** to help passengers locate gates and facilities easily.
- Reduce **waiting time, queues, and dependency on manual airport services**.
- Provide convenient access to **passenger assistance and airport facilities**.
- Ensure a **secure, user-friendly, and responsive** mobile experience.
- Improve the overall passenger journey through a **seamless, personalized, and connected self-service system**.

Modules

1. **User Authentication & Profile**
   - Sign up, login, OTP verification, and passenger profile management.
2. **Flight & Booking Management**
   - Search flights, view booking/PNR details, and check real-time flight status.
3. **Self Check-in & Boarding Pass**
   - Digital check-in, seat selection, and QR-based boarding pass generation.
4. **Baggage Tracking**
   - Track baggage status from check-in to baggage collection.
5. **Airport Navigation**
   - Interactive airport map with gates, terminals, security, lounges, restaurants, and other facilities.
6. **Notifications & Alerts**
   - Real-time updates for gate changes, delays, boarding, cancellations, and baggage status.
7. **Passenger Services**
   - Special assistance, wheelchair requests, lost & found, parking, transport, and airport information.
8. **Admin & Database Management**
   - Manage passengers, flights, bookings, baggage, airport facilities, notifications, and service requests.

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SOFTWARE REQUIREMENTS

Frontend / Mobile App

- React Native
- Expo
- JavaScript / TypeScript

Backend

- Node.js
- Express.js
- REST API

Database

- MySQL
- MySQL Workbench

Development Tools

- Visual Studio Code
- Android Studio
- Git & GitHub

APIs & Libraries

- Google Maps API – airport navigation
- JWT – authentication
- QR Code Library – digital boarding pass
- Axios – API communication

UI/UX & Testing

- Figma – interface design
- Postman – API testing

Deployment

- Expo / EAS – mobile app build
- Render – backend deployment

Operating System

- Windows 10/11
- Android/iOS for application testing

CONCLUSION AND FUTURE SCOPE

Conclusion

The **Airline & Airport Mobile App** provides a single platform for check-in, boarding passes, flight updates, baggage tracking, airport navigation, and passenger assistance. It aims to **reduce waiting time and improve the overall travel experience**.

Future Scope

- AI-based travel assistance
- Real-time indoor airport navigation
- Biometric and contactless services
- Smart baggage tracking
- Multilingual voice assistance
- Integration with more airlines and airports

THANK YOU